THE FINITE FREE STAM INEQUALITY: STRUCTURAL IDENTITIES, THE GAUSSIAN-INPUT CASE, AND REDUCTION PRINCIPLES

Abstract. Let $\mathcal{P}_n^{\mathbb{R}}$ denote the set of monic, degree-n, real-rooted polynomials and let $\boxplus_n$ be the Marcus–Spielman–Srivastava finite free additive convolution. For $r \in \mathcal{P}_n^{\mathbb{R}}$ with simple roots $\lambda_1 < \cdots < \lambda_n$, the *finite free Fisher information* is $\Phi_n(r) := \sum_{i=1}^n V_i^2$, where $V_i := \sum_{j \neq i} (\lambda_i - \lambda_j)^{-1}$. We prove the *finite free Stam inequality*

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}, \qquad p, q \in \mathcal{P}_n^{\mathbb{R}},$$

for all $n \geq 2$, confirming a conjecture of D. Shlyakhtenko. The proof, due to Garza Vargas, Srivastava, and Stier [7], proceeds in three steps: (1) a Jacobian identity relating score vectors under convolution, derived from the commutation of $\boxplus_n$ with differential operators; (2) a contraction inequality for the Jacobian of the root map, established via variance additivity and a convexity result of Bauschke et al. [1] on hyperbolic polynomials; (3) Blachman's classical argument adapted to the finite free setting. We also develop structural identities ($\Phi_n = 2\mathcal{R} = \mathrm{tr}(L)$, Bezoutian representation), the Hermite heat equation and root ODE, and give independent proofs for $n \leq 3$ and for Gaussian inputs at all n.

1. Introduction

The classical Stam inequality [11] states that for independent continuous random variables X, Y with finite Fisher informations $J(X), J(Y)$:

$$\frac{1}{J(X+Y)} \geq \frac{1}{J(X)} + \frac{1}{J(Y)}. \tag{1}$$

This is a cornerstone of information theory, closely related to the entropy power inequality [3] and the Cramér–Rao bound (see [4] for a survey).

Marcus, Spielman, and Srivastava [9, 8] introduced the *finite free additive convolution* $\boxplus_n$ on monic real-rooted polynomials of degree n, a finite-dimensional analogue of free additive convolution in the sense of [12]. A natural question is whether the Stam inequality (1) has a polynomial analogue. Define the *finite free Fisher information* of $r \in \mathcal{P}_n^{\mathbb{R}}$ with simple roots $\lambda_1 < \cdots < \lambda_n$ by $\Phi_n(r) := \sum_{i=1}^n (\sum_{j \neq i} (\lambda_i - \lambda_j)^{-1})^2$. The *finite free Stam inequality*, conjectured by D. Shlyakhtenko and proved by Garza Vargas, Srivastava, and Stier [7], states that for all $p, q \in \mathcal{P}_n^{\mathbb{R}}$:

$$\frac{1}{\Phi_n(p \boxplus_n q)} \geq \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}. \tag{2}$$

Preserving real-rootedness under $\boxplus_n$ is guaranteed by [9]; see [2] for the connection to linear operators preserving stability.

Contributions.

(i) Full proof of the Stam inequality for all $n \geq 2$ (Section 5), following [7].
(ii) Structural identities (Section 3): $\Phi_n = 2\mathcal{R} = \mathrm{tr}(L)$, score identities, variance additivity, Bezoutian and Laplacian formulations.
(iii) The Hermite heat equation and root ODE (Section 2); independent proofs for $n \leq 3$ and Gaussian-input Stam at all n (Section 4).

Roadmap. Section 2 establishes notation and the key foundational results: the heat equation for the Hermite flow (Theorem 3.9) and the root ODE (Corollary 3.11). Section 3 develops the Fisher–repulsion, Laplacian, and Bezoutian identities. Section 4 gives independent proofs for small cases ($n \leq 3$, Gaussian input). Section 5 presents the full proof for all n. Section 6 concludes.

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2. Preliminaries

Definition 2.1 (MSS convolution [9])**.** For $p(x) = \sum_{k=0}^{n} a_k x^{n-k}$ and $q(x) = \sum_{k=0}^{n} b_k x^{n-k}$ with $a_0 = b_0 = 1$, the *finite free additive convolution* $r = p \boxplus_n q$ is defined by $r(x) = \sum_{k=0}^{n} c_k x^{n-k}$ with

$$c_k = \sum_{i+j=k} \frac{(n-i)!\,(n-j)!}{n!\,(n-k)!}\, a_i b_j.$$

By [9], $\boxplus_n$ preserves $\mathcal{P}_n^{\mathbb{R}}$.

Definition 2.2 (K-transform and log-cumulants)**.** Define

$$\kappa_k(r) := \frac{(n-k)!\,c_k(r)}{n!}, \qquad K_r(z) := \sum_{k=0}^{n} \kappa_k(r)\, z^k.$$

Then

$$K_{p \boxplus_n q}(z) = K_p(z) \cdot K_q(z) \pmod{z^{n+1}}.$$

The *log-cumulants* $\ell_k(r) := [z^k] \log K_r(z)$ satisfy $\ell_1 = \kappa_1$ and $\ell_k = \kappa_k - \frac{1}{k}\sum_{j=1}^{k-1} j\,\ell_j \kappa_{k-j}$ for $k \geq 2$. They are **additive**: $\ell_k(p \boxplus_n q) = \ell_k(p) + \ell_k(q)$ for all k.

Definition 2.3 (Scores and Fisher information)**.** For $r \in \mathcal{P}_n^{\mathbb{R}}$ with simple roots $\lambda_1 < \cdots < \lambda_n$, let $V_i(r) := \sum_{j \neq i} (\lambda_i - \lambda_j)^{-1}$ (the *score vector* $V = (V_1, \ldots, V_n)$), $\Phi_n(r) := \sum_i V_i^2$ (the *Fisher information*), $\mathcal{R}(r) := \sum_{i<j} (\lambda_i - \lambda_j)^{-2}$ (the *repulsion energy*), and $\mathcal{S}(r) := \sum_{i<j} (V_i - V_j)^2/(\lambda_i - \lambda_j)^2$ (the *score-gradient energy*). If r has a repeated root, set $\Phi_n(r) = \infty$.

Definition 2.4 (Graph Laplacian)**.** The graph Laplacian of r is $L \in \mathbb{R}^{n \times n}$ with $L_{ij} = -(\lambda_i - \lambda_j)^{-2}$ for $i \neq j$ and $L_{ii} = \sum_{k \neq i} (\lambda_i - \lambda_k)^{-2}$. We have $L\mathbf{1} = 0$, $L \succeq 0$, $\operatorname{rank} L = n-1$. Equivalently, $L = -\frac{1}{2}\mathrm{Hess}_\lambda(\log \operatorname{disc}(r))$.

Definition 2.5 (Variance and Gaussian polynomials)**.** For $r \in \mathcal{P}_n^{\mathbb{R}}$: $\mu(r) := n^{-1}\sum_i \lambda_i$, $\sigma^2(r) := n^{-1}\sum_i (\lambda_i - \mu)^2$. Both are additive under $\boxplus_n$ (Lemma 3.7). The *additive variance parameter* $u := \sigma^2/(2(n-1))$ satisfies $u(p \boxplus_n q) = u(p) + u(q)$. The *finite Gaussian* $g_t \in \mathcal{P}_n^{\mathbb{R}}$ has $\sigma^2(g_t) = t$ and $\ell_k(g_t) = 0$ for $k \geq 3$. The Hermite semigroup satisfies $g_s \boxplus_n g_t = g_{s+t}$.

Definition 2.6 (Normalised cumulant ratios)**.** For centred $r \in \mathcal{P}_n^{\mathbb{R}}$ with $u := -\ell_2(r) > 0$, define $\tau_k(r) := \ell_k(r)/u(r)^{k/2}$ for $k \geq 3$.

Lemma 2.7 (Normalisation identities)**.** *For centred $r \in \mathcal{P}_n^{\mathbb{R}}$ (i.e., $\mu(r) = 0$), the parameters κ_2, ℓ_2, u, and σ^2 are related by:*

$$\ell_2 = \kappa_2 = \frac{(n-2)!\,a_2}{n!} = \frac{a_2}{n(n-1)}, \qquad u := -\ell_2 > 0, \qquad \sigma^2 = 2(n-1)\,u. \tag{3}$$

Here a_2 is the coefficient of x^{n-2} in r (so $a_2 < 0$ for centred real-rooted r with $n \geq 2$).

Proof. From Definition 2.2: $\kappa_2 = (n-2)!\,a_2/n!$. The log-cumulant recurrence (Definition 2.2) gives $\ell_2 = \kappa_2 - \frac{1}{2}\kappa_1^2 = \kappa_2$ when r is centred ($\kappa_1 = \ell_1 = 0$). From the variance formula with $a_1 = 0$: $\sigma^2 = -2a_2/n = -2n(n-1)\ell_2/n = 2(n-1)(-\ell_2) = 2(n-1)u$. All three parameters are additive under $\boxplus_n$ because ℓ_2 is additive (Definition 2.2). □

3. Structural identities

We collect the main identities connecting Φ_n to spectral quantities. Throughout this section, $r \in \mathcal{P}_n^{\mathbb{R}}$ has simple roots.

Theorem 3.1 (Fisher–repulsion identity)**.** $\Phi_n(r) = 2\,\mathcal{R}(r)$.

Proof. Expand $\Phi_n = \sum_i V_i^2 = \sum_i \sum_{j \neq i} \sum_{k \neq i} (\lambda_i - \lambda_j)^{-1}(\lambda_i - \lambda_k)^{-1}$. The diagonal terms ($j = k$) sum to $2\sum_{i<j}(\lambda_i - \lambda_j)^{-2} = 2\mathcal{R}$. The cross-terms ($j \neq k$, both $\neq i$) group into triples $\{a, b, c\}$, each contributing $(a-b)^{-1}(a-c)^{-1} + (b-a)^{-1}(b-c)^{-1} + (c-a)^{-1}(c-b)^{-1} = 0$ by the partial-fraction identity. □

Theorem 3.2 (Fisher–Laplacian identities)**.** (a) $\Phi_n = \operatorname{tr}(L)$.
(b) $V = L\lambda$ (Euler identity).
(c) $\lambda^T L \lambda = \binom{n}{2}$.
(d) $\Phi_n = \|L\lambda\|^2 = \lambda^T L^2 \lambda$.

Proof. (a) $\operatorname{tr}(L) = \sum_i \sum_{k \neq i} (\lambda_i - \lambda_k)^{-2} = 2\sum_{i<j}(\lambda_i - \lambda_j)^{-2} = 2\mathcal{R} = \Phi_n$.
(b) $(L\lambda)_i = \sum_{j \neq i} (\lambda_i - \lambda_j)/(\lambda_i - \lambda_j)^2 = \sum_{j \neq i} (\lambda_i - \lambda_j)^{-1} = V_i$.
(c) $\lambda^T L \lambda = V \cdot \lambda = \sum_i \lambda_i V_i = \binom{n}{2}$ by the Euler identity for disc (degree $n(n-1)$ homogeneous).
(d) Immediate from $V = L\lambda$. □

Lemma 3.3 (Score identities). (i) $\sum_i V_i = 0$.
(ii) $\sum_i (\lambda_i - \mu)V_i = \binom{n}{2}$.
(iii) $\Phi_n = \sum_{i<j}(V_i - V_j)/(\lambda_i - \lambda_j)$.
(iv) $V_i = r''(\lambda_i)/(2r'(\lambda_i))$.

Proof. (i) $\sum_i V_i = \sum_{i\neq j}(\lambda_i - \lambda_j)^{-1} = 0$ (antisymmetric).

(ii) $\sum_i \lambda_i V_i = \sum_{i\neq j}\lambda_i/(\lambda_i - \lambda_j) = \sum_{i\neq j}\big[1 + \lambda_j/(\lambda_i - \lambda_j)\big] = n(n-1) + \sum_{i\neq j}\lambda_j/(\lambda_i - \lambda_j)$. Using $\sum_{i\neq j}\lambda_j/(\lambda_i - \lambda_j) = -\sum_{i\neq j}\lambda_i/(\lambda_j - \lambda_i) = -\sum_i \lambda_i V_i$, we get $2\sum_i \lambda_i V_i = n(n-1)$, so $\sum_i \lambda_i V_i = \binom{n}{2}$. By (i), subtracting $\mu\sum V_i = 0$ gives (ii).

(iii) Expand the right-hand side:

$$\sum_{i<j}\frac{V_i - V_j}{\lambda_i - \lambda_j} = \sum_{i<j}\frac{1}{\lambda_i - \lambda_j}\Big(\sum_{k\neq i}\frac{1}{\lambda_i - \lambda_k} - \sum_{k\neq j}\frac{1}{\lambda_j - \lambda_k}\Big)$$
$$= \sum_{i<j}\sum_{k\neq i}\frac{1}{(\lambda_i - \lambda_j)(\lambda_i - \lambda_k)} - \sum_{i<j}\sum_{k\neq j}\frac{1}{(\lambda_i - \lambda_j)(\lambda_j - \lambda_k)}.$$

Relabelling $i \leftrightarrow j$ in the second sum and combining yields $2\sum_{i<j}\sum_{k\neq i} 1/((\lambda_i - \lambda_j)(\lambda_i - \lambda_k))$. Separating diagonal ($k = j$) from cross ($k \neq i, j$) terms: the diagonal gives $\sum_i \sum_{j\neq i}(\lambda_i - \lambda_j)^{-2} = \Phi_n$; the cross-terms group into triples $\{i, j, k\}$, each contributing $\sum_{\text{cyc}} 1/((a-b)(a-c)) = 0$ by the same partial-fraction identity as in Theorem 3.1.

(iv) Since $r'(\lambda_i) = \prod_{j\neq i}(\lambda_i - \lambda_j)$, we have $V_i = r''(\lambda_i)/(2r'(\lambda_i))$. □

Theorem 3.4 (Fisher–variance inequality). $\Phi_n(r)\cdot\sigma^2(r) \geq n(n-1)^2/4$.

Proof. Cauchy–Schwarz on $\sum_i(\lambda_i - \mu)V_i = \binom{n}{2}$ with $\sum V_i = 0$: $\big|\sum(\lambda_i - \mu)V_i\big|^2 \leq \big(\sum(\lambda_i - \mu)^2\big)\big(\sum V_i^2\big) = n\sigma^2\cdot\Phi_n$. Hence $n\sigma^2\cdot\Phi_n \geq \binom{n}{2}^2 = n^2(n-1)^2/4$. □

Theorem 3.5 (Score-gradient inequality). $\mathcal{S}(r)\cdot\sigma^2(r) \geq (n-1)\Phi_n(r)/2$.

Proof. Write $\lambda_c := \lambda - \mu\mathbf{1}$ for the centred root vector. Since $L\mathbf{1} = 0$, $V = L\lambda = L\lambda_c$. The Cauchy–Schwarz inequality for the positive semi-definite form $\langle u, v\rangle_L := u^T L v$ gives $(\lambda_c^T L^2\lambda_c)^2 \leq (\lambda_c^T L\lambda_c)(\lambda_c^T L^3\lambda_c)$, i.e., $\Phi_n^2 \leq \binom{n}{2}\cdot\mathcal{S}$. Combining with the Fisher–variance inequality (Theorem 3.4):

$$\mathcal{S}\,\sigma^2 \;\geq\; \frac{\Phi_n^2}{\binom{n}{2}}\,\sigma^2 \;=\; \frac{\Phi_n\,\sigma^2}{\binom{n}{2}}\cdot\Phi_n \;\geq\; \frac{n(n-1)^2/4}{n(n-1)/2}\cdot\Phi_n \;=\; \frac{(n-1)\,\Phi_n}{2}.$$ □

Theorem 3.6 (Bezoutian representation). $\Phi_n(r) = \sum_{i=1}^n r''(\lambda_i)^2/(4\,r'(\lambda_i)^2) = \|r''/2\|^2_{\mathrm{Bez}(r,r')}$.

Proof. The Bezoutian matrix $\mathrm{Bez}(r, r')$ is the unique symmetric $B \in \mathbb{R}^{n\times n}$ satisfying $\sum_{i,j} B_{ij}x^{n-1-i}y^{n-1-j} = \big(r(x)r'(y) - r'(x)r(y)\big)/(x - y)$. The associated inner product is diagonal in the Lagrange basis $\{L_i(x) = \prod_{j\neq i}(x - \lambda_j)/\prod_{j\neq i}(\lambda_i - \lambda_j)\}$: $\langle f, g\rangle_{\mathrm{Bez}} = \sum_i f(\lambda_i)g(\lambda_i)/r'(\lambda_i)^2$ (see [6] for the diagonalisation). Since $V_i = r''(\lambda_i)/(2r'(\lambda_i))$ (Lemma 3.3(iv)), we get $\Phi_n = \sum V_i^2 = \|r''/2\|^2_{\mathrm{Bez}}$. □

Lemma 3.7 (Variance additivity). $\sigma^2(p \boxplus_n q) = \sigma^2(p) + \sigma^2(q)$.

Proof. From the MSS coefficient formula (Definition 2.1): $c_1 = a_1 + b_1$, $c_2 = a_2 + b_2 + \frac{n-1}{n}a_1b_1$. Using $\sigma^2 = \frac{(n-1)a_1^2 - 2na_2}{n^2}$:

$$\sigma^2(p \boxplus_n q) = \frac{(n-1)(a_1+b_1)^2 - 2n(a_2+b_2+\frac{n-1}{n}a_1b_1)}{n^2}$$
$$= \frac{(n-1)a_1^2 - 2na_2}{n^2} + \frac{(n-1)b_1^2 - 2nb_2}{n^2} + \frac{2(n-1)a_1b_1 - 2(n-1)a_1b_1}{n^2}$$
$$= \sigma^2(p) + \sigma^2(q).$$ □

Lemma 3.8 (Derivative compatibility). $(p \boxplus_n q)'/n = (p'/n) \boxplus_{n-1} (q'/n)$.

Proof. The monic degree-$(n-1)$ polynomial p'/n has coefficients $\tilde{a}_k = (n-k)a_k/n$. A direct calculation confirms compatibility of the $\boxplus_{n-1}$ formula with differentiation, using $(n-i)(n-j)/(n^2)\cdot(n-1-i)!(n-1-j)!/((n-1)!(n-1-k)!) = (n-k)/n\cdot(n-i)!(n-j)!/(n!(n-k)!)$ for $i + j = k$. □

Theorem 3.9 (Hermite heat equation). *The Hermite flow $r_t := r \boxplus_n g_t$ satisfies the scaled backwards heat equation*

$$\partial_t r_t(x) \;=\; -\frac{1}{2(n-1)}\,r_t''(x), \tag{4}$$

with initial condition $r_0 = r$.

Proof. The K-transform of the finite Gaussian is $K_{g_t}(z) = \exp\bigl(-\frac{t}{2(n-1)}z^2\bigr) \bmod z^{n+1}$, since $\ell_2(g_t) = -t/(2(n-1))$ and $\ell_k(g_t) = 0$ for $k \geq 3$ (Definition 2.5). Hence $K_{r_t}(z) = K_r(z)\,K_{g_t}(z)$ gives $\partial_t K_{r_t}(z) = -\frac{z^2}{2(n-1)}\,K_{r_t}(z) \bmod z^{n+1}$. Expanding $K_{r_t}(z) = \sum_{k=0}^{n} \kappa_k(t)\,z^k$ yields the coefficient ODE

$$\kappa_k'(t) = -\frac{1}{2(n-1)}\,\kappa_{k-2}(t) \quad (k \geq 2), \qquad \kappa_0' = \kappa_1' = 0.$$

Translating to the standard coefficients $c_k(t) = n!\,\kappa_k(t)/(n-k)!$:

$$c_k'(t) = -\frac{(n-k+2)(n-k+1)}{2(n-1)}\,c_{k-2}(t) \qquad (k \geq 2).$$

Now compute $\partial_t r_t(x) = \sum_{k=0}^{n} c_k'(t)\,x^{n-k}$ and observe that $r_t''(x) = \sum_{j=0}^{n-2}(n-j)(n-j-1)\,c_j(t)\,x^{n-j-2}$; substituting $k = j+2$ identifies the two sums, giving (4). □

Remark 3.10 (Context). The heat equation (4) is exact for polynomials of degree n (no truncation is involved, since all coefficients lie in degrees $\leq n$). The root ODE (5) is a deterministic analogue of the Dyson Brownian motion for eigenvalues of random Hermitian matrices [5]; here it arises purely algebraically from the K-transform.

Corollary 3.11 (Root ODE)**.** *Along the Hermite flow $r_t = r \boxplus_n g_t$, each simple root $\lambda_i(t)$ of r_t satisfies*

$$\dot\lambda_i(t) \;=\; \frac{V_i(t)}{n-1} \;=\; \frac{1}{n-1}\sum_{j\neq i}\frac{1}{\lambda_i(t)-\lambda_j(t)}. \tag{5}$$

Proof. Differentiating $r_t(\lambda_i(t)) = 0$ in t and applying Theorem 3.9:

$$0 = \partial_t r_t(\lambda_i) + r_t'(\lambda_i)\,\dot\lambda_i = -\frac{r_t''(\lambda_i)}{2(n-1)} + r_t'(\lambda_i)\,\dot\lambda_i,$$

so $\dot\lambda_i = r_t''(\lambda_i)/(2(n-1)\,r_t'(\lambda_i)) = V_i/(n-1)$ by Lemma 3.3(iv). □

Theorem 3.12 (De Bruijn identity)**.** *Along the Hermite flow $r_t = r \boxplus_n g_t$: $\frac{d}{dt}\log|\operatorname{disc}(r_t)| = \frac{2}{n-1}\Phi_n(r_t)$.*

Proof. By Corollary 3.11, $\dot\lambda_i = V_i/(n-1)$. Since $\operatorname{disc}(r) = \prod_{i<j}(\lambda_i-\lambda_j)^2$, we have $\partial_{\lambda_i}\log\operatorname{disc} = 2V_i$. Therefore $\frac{d}{dt}\log\operatorname{disc} = \sum_i 2V_i\cdot V_i/(n-1) = 2\Phi_n/(n-1)$. □

4. Proved cases of the Stam inequality

4.1. The case $n = 2$: equality.

For $n = 2$: $\Phi_2(r) = 2/(\lambda_1-\lambda_2)^2 = 1/(2\sigma^2)$, so $1/\Phi_2 = 2\sigma^2$. The Stam inequality reduces to variance additivity (Lemma 3.7), with equality.

4.2. The case $n = 3$: SOS proof.

Theorem 4.1 (Stam for $n = 3$)**.** *For centred $p, q \in \mathcal{P}_3^{\mathbb{R}}$ with $u_p, u_q > 0$, let $r = p \boxplus_n q$, $w = u_p/(u_p+u_q)$, $\alpha := \ell_3(p)/u_p$, $\beta := \ell_3(q)/u_q$. Then*

$$D_3 := \frac{1}{\Phi_3(r)} - \frac{1}{\Phi_3(p)} - \frac{1}{\Phi_3(q)} = \frac{3}{2}\bigl[(1-w)\alpha^2 + w(1-w)(\alpha-\beta)^2 + w\beta^2\bigr] \geq 0. \tag{6}$$

Equality holds (for $w \in (0,1)$) iff $\ell_3(p) = \ell_3(q) = 0$.

Proof. Step 1: Log-cumulants for the depressed cubic. Let $r(x) = x^3 + e_2x + e_3$ be a centred monic cubic. The K-transform coefficients (Definition 2.2) are $\kappa_0 = 1$, $\kappa_1 = 0$, $\kappa_2 = \frac{1!\cdot e_2}{3!} = \frac{e_2}{6}$, $\kappa_3 = \frac{0!\cdot e_3}{3!} = \frac{e_3}{6}$, so $K_r(z) = 1 + \frac{e_2}{6}z^2 + \frac{e_3}{6}z^3$. Since $\log(1+x) = x - \frac{x^2}{2} + \cdots$ and K_r has no z^1 term: $\ell_2 = [z^2]\log K_r = \kappa_2 = e_2/6$, $\ell_3 = [z^3]\log K_r = \kappa_3 = e_3/6$. (The cross-term $\kappa_2^2 z^4/2$ contributes to $[z^4]\log K_r$, not to $[z^3]$.) Hence $u := -\ell_2 = -e_2/6 > 0$ and $e_2 = -6u$, $e_3 = 6\ell_3$.

Step 2: Φ_3 via the repulsion energy. Denote the roots $\lambda_1 < \lambda_2 < \lambda_3$ and the three squared gaps $D_{ij} := (\lambda_i-\lambda_j)^2$. Since $r'(\lambda_i) = \prod_{j\neq i}(\lambda_i-\lambda_j)$, we have $r'(\lambda_i)^2 = \prod_{j\neq i} D_{ij}$. Hence the sum of products of two squared gaps equals $\sum_i r'(\lambda_i)^2$. For $r'(x) = 3x^2 + e_2$ and the Newton power sums $p_2 = -2e_2$, $p_4 = 2e_2^2$:

$$\sum_{i=1}^{3} r'(\lambda_i)^2 = \sum_i (3\lambda_i^2 + e_2)^2 = 9\,p_4 + 6\,e_2\,p_2 + 3\,e_2^2 = 18e_2^2 - 12e_2^2 + 3e_2^2 = 9e_2^2.$$

The discriminant is $\Delta_3 = \prod_{i<j} D_{ij} = -4e_2^3 - 27e_3^2 = 864\,u^3 - 972\,\ell_3^2$, and the repulsion energy decomposes as

$$\mathcal{R} = \sum_{i<j}\frac{1}{D_{ij}} = \frac{\sum_i r'(\lambda_i)^2}{\Delta_3} = \frac{9e_2^2}{\Delta_3}.$$

By Theorem 3.1, $\Phi_3 = 2\mathcal{R} = 18e_2^2/\Delta_3 = 648\,u^2/(864\,u^3 - 972\,\ell_3^2)$.

Step 3: Closed-form reciprocal.

$$\frac{1}{\Phi_3(r)} = \frac{864\,u^3 - 972\,\ell_3^2}{648\,u^2} = \frac{4u}{3} - \frac{3\ell_3^2}{2u^2}, \tag{7}$$

where the last equality uses $864/648 = 4/3$ and $972/648 = 3/2$.

Step 4: Defect computation. Since u and ℓ_3 are additive under $\boxplus_n$, set $u_r = u_p + u_q$ and $\ell_{3,r} = \ell_{3,p} + \ell_{3,q}$. With $\alpha = \ell_{3,p}/u_p$ and $\beta = \ell_{3,q}/u_q$:

$$\begin{aligned} D_3 &= \frac{4u_r}{3} - \frac{3\ell_{3,r}^2}{2u_r^2} - \frac{4u_p}{3} + \frac{3\ell_{3,p}^2}{2u_p^2} - \frac{4u_q}{3} + \frac{3\ell_{3,q}^2}{2u_q^2} \\ &= \frac{3}{2}\left[\alpha^2 + \beta^2 - \frac{(u_p\alpha + u_q\beta)^2}{(u_p+u_q)^2}\right], \end{aligned}$$

since the linear terms $\frac{4}{3}(u_r - u_p - u_q) = 0$ cancel by additivity. Writing $w = u_p/(u_p+u_q)$:

$$\begin{aligned} D_3 &= \frac{3}{2}\big[\alpha^2 + \beta^2 - w^2\alpha^2 - 2w(1-w)\alpha\beta - (1-w)^2\beta^2\big] \\ &= \frac{3}{2}\big[(1-w^2)\alpha^2 - 2w(1-w)\alpha\beta + w(2-w)\beta^2\big]. \end{aligned}$$

Since $(1-w)(1+w) = 1-w^2$ and $w(2-w) = w(1-w) + w$:

$$D_3 = \frac{3}{2}\big[(1-w)\alpha^2 + w(1-w)(\alpha-\beta)^2 + w\beta^2\big] \geq 0.$$

Each of the three summands is manifestly non-negative; for $w \in (0,1)$, all vanish iff $\alpha = \beta = 0$, i.e. $\ell_3(p) = \ell_3(q) = 0$. □

Remark 4.2. The structure of $1/\Phi_3$ is $1/\Phi_3 = A(u) + Q(\ell_3/u)$ where $A(u) = 4u/3$ is additive under $\boxplus_n$ and $Q(\cdot) = -\frac{3}{2}(\cdot)^2$ is concave. The Stam defect therefore reduces to the concavity defect of a quadratic composed with a weighted-linear mixing law. The Hessian of $1/\Phi_3$ in (u, ℓ_3)-coordinates is **not** negative semi-definite, so the result does not follow from global concavity of $1/\Phi_3$ as a function of both variables.

4.3. Gaussian-input Stam for all n.

Theorem 4.3 (Gaussian-input Stam inequality)**.** *For all $r \in \mathcal{P}_n^{\mathbb{R}}$ and $t > 0$: $1/\Phi_n(r \boxplus_n g_t) \geq 1/\Phi_n(r) + 1/\Phi_n(g_t)$. Equality holds on the Hermite manifold.*

Proof. Step 1: Gaussian Fisher information. The finite Gaussian g_t has roots at the n zeros of the probabilist Hermite polynomial He_n scaled so that $\sigma^2(g_t) = t$ and $\ell_k(g_t) = 0$ for $k \geq 3$. The Hermite differential equation $\mathrm{He}_n''(x) - x\,\mathrm{He}_n'(x) + n\,\mathrm{He}_n(x) = 0$ at a root λ_i (where $\mathrm{He}_n(\lambda_i) = 0$) gives $\mathrm{He}_n''(\lambda_i) = \lambda_i\,\mathrm{He}_n'(\lambda_i)$, hence $V_i = \mathrm{He}_n''(\lambda_i)/(2\mathrm{He}_n'(\lambda_i)) = \lambda_i/2$. After scaling, $V_i(g_t) = c\,(\lambda_i - \mu)$ for a constant c depending only on n and t, so equality holds in the Fisher–variance inequality (Theorem 3.4): $\Phi_n(g_t) = n(n-1)^2/(4t)$ and $1/\Phi_n(g_t) = 4t/(n(n-1)^2)$.

Step 2: Root ODE. Let $r_t := r \boxplus_n g_t$ with roots $\lambda_1(t) < \cdots < \lambda_n(t)$. By Corollary 3.11, each root satisfies $\dot\lambda_i = V_i/(n-1)$, where $V_i = V_i(r_t)$.

Step 3: Score ODE. Differentiating $V_i = \sum_{j\neq i}(\lambda_i - \lambda_j)^{-1}$ with respect to t:

$$\dot V_i = -\sum_{j\neq i}\frac{\dot\lambda_i - \dot\lambda_j}{(\lambda_i-\lambda_j)^2} = -\frac{1}{n-1}\sum_{j\neq i}\frac{V_i - V_j}{(\lambda_i-\lambda_j)^2} = -\frac{(LV)_i}{n-1},$$

where L is the graph Laplacian (Definition 2.4).

Step 4: Fisher dissipation. Define the *score-gradient energy* $\mathcal{S} := V^T L V = \sum_{i<j}(V_i - V_j)^2/(\lambda_i - \lambda_j)^2$. Then $\frac{d}{dt}\Phi_n(r_t) = 2\sum_i V_i\dot V_i = -\frac{2}{n-1}V^T L V = -\frac{2}{n-1}\mathcal{S}$. Since $L \succeq 0$, we have $\mathcal{S} \geq 0$, hence $\Phi_n'(r_t) \leq 0$.

Step 5: Lower bound on $\mathcal{S}/\Phi_n^2$. Since $L\mathbf{1} = 0$, $V = L\lambda = L\lambda_c$ where $\lambda_c := \lambda - \mu\mathbf{1}$. Hence $\Phi_n = V^T V = \lambda_c^T L^2 \lambda_c$ and $\mathcal{S} = V^T L V = \lambda_c^T L^3 \lambda_c$. The Cauchy–Schwarz inequality for the semi-definite form $\langle u, v\rangle_L = u^T L v$ gives $(\lambda_c^T L^2 \lambda_c)^2 \leq (\lambda_c^T L \lambda_c)(\lambda_c^T L^3 \lambda_c) = \binom{n}{2}\mathcal{S}$, hence $\Phi_n^2 \leq \binom{n}{2}\mathcal{S}$, i.e.,

$$\frac{\mathcal{S}}{\Phi_n^2} \geq \frac{1}{\binom{n}{2}} = \frac{2}{n(n-1)}.$$

Step 6: Integration. Since $\big(1/\Phi_n\big)' = -\Phi_n'/(\Phi_n^2) = \frac{2}{n-1}\cdot\frac{\mathcal{S}}{\Phi_n^2} \geq \frac{2}{n-1}\cdot\frac{2}{n(n-1)} = \frac{4}{n(n-1)^2}$, integrating from 0 to t:

$$\frac{1}{\Phi_n(r_t)} - \frac{1}{\Phi_n(r)} \geq \frac{4t}{n(n-1)^2} = \frac{1}{\Phi_n(g_t)}.$$

Equality saturation. When $r = g_s$ is itself a finite Gaussian, $r_t = g_s \boxplus_n g_t = g_{s+t}$, and the scores satisfy $V_i(g_u) = c(\lambda_i - \mu)$ for every $u > 0$ (Step 1). Then $\mathcal{S}/\Phi_n^2 = 1/\binom{n}{2}$ exactly, so $(1/\Phi_n)' = 4/(n(n-1)^2)$ for all $t > 0$, and the integrated bound is attained with equality. □

5. Proof of the Stam inequality for all n

We now present the full proof of the finite free Stam inequality, following Garza Vargas, Srivastava, and Stier [7]. The argument uses three ingredients: (1) a relation between score vectors mediated by the Jacobian of the root map; (2) a contraction inequality for this Jacobian, derived from variance additivity and a convexity result on hyperbolic polynomials; (3) Blachman's classical argument [3].

5.1. The root map and its Jacobian.

Definition 5.1 (Root map)**.** For $p, q \in \mathcal{P}_n^{\mathbb{R}}$ with root vectors $\alpha = (\alpha_1 \geq \cdots \geq \alpha_n)$ and $\beta = (\beta_1 \geq \cdots \geq \beta_n)$, define $\Omega_{\boxplus_n}(\alpha, \beta) \in \mathbb{R}^n$ to be the ordered root vector of $p \boxplus_n q$. By [9], $\Omega_{\boxplus_n}$ is well-defined and takes values in *ordered* real vectors. Write $\gamma = \Omega_{\boxplus_n}(\alpha, \beta)$. Let $J_{\boxplus_n}$ denote the Jacobian (the $n \times 2n$ matrix of partial derivatives) of $\Omega_{\boxplus_n}$ at (α, β), defined whenever the roots are simple.

Lemma 5.2 (Approximation by simple roots)**.** *Let $T_\varepsilon := (1 - \varepsilon\, d/dx)^n$. For any monic real-rooted p of degree n:*

(a) *$T_\varepsilon p$ is monic, real-rooted of degree n, with simple roots (Nuij [10]).*
(b) *$\Phi_n(T_\varepsilon p) \to \Phi_n(p)$ as $\varepsilon \to 0$.*
(c) *$(T_\varepsilon p) \boxplus_n (T_\varepsilon q) = T_\varepsilon^2(p \boxplus_n q)$, since $\boxplus_n$ commutes with differential operators [9].*

Hence it suffices to prove Theorem 5.7 when p, q, and $p \boxplus_n q$ all have simple roots.

5.2. Score vectors and the Jacobian.

Proposition 5.3 (Score–Jacobian relation)**.** *Let $p, q \in \mathcal{P}_n^{\mathbb{R}}$ have simple roots with root vectors α, β and $\gamma = \Omega_{\boxplus_n}(\alpha, \beta)$. For any $a, b \geq 0$:*

$$J_{\boxplus_n} \begin{pmatrix} a\, V(\alpha) \\ b\, V(\beta) \end{pmatrix} = (a+b)\, V(\gamma), \tag{8}$$

where $V(\alpha) = \left(\sum_{j \neq i} (\alpha_i - \alpha_j)^{-1}\right)_{i=1}^n$ is the score vector.

Proof. For $t \geq 0$, define $p_t := \exp(-\frac{t}{2}\partial_x^2)\, p$ with ordered root vector $\alpha(t)$, and similarly q_t, r_t with $\beta(t), \gamma(t)$. Since $\boxplus_n$ commutes with any differential operator, $r_{(a+b)t} = p_{at} \boxplus_n q_{bt}$, whence $\gamma((a+b)t) = \Omega_{\boxplus_n}(\alpha(at), \beta(bt))$. Differentiating at $t = 0$ and using the chain rule:

$$(a+b)\, \gamma'(0) = J_{\boxplus_n} \begin{pmatrix} a\, \alpha'(0) \\ b\, \beta'(0) \end{pmatrix}.$$

By Lemma 3.3(iv), $\alpha_i'(0) = p''(\alpha_i)/(2p'(\alpha_i)) = V_i(\alpha)$ (the same computation as in Corollary 3.11). □

5.3. Jacobian contraction.

Proposition 5.4 (Jacobian contraction on V)**.** *Let $V := \{(u, v) \in \mathbb{R}^n \times \mathbb{R}^n : u^T \mathbf{1} = v^T \mathbf{1} = 0\}$. For all $(u, v) \in V$:*

$$\left\| J_{\boxplus_n}(u, v) \right\|^2 \;\leq\; \|u\|^2 + \|v\|^2. \tag{9}$$

The proof of this proposition requires two intermediate results.

Lemma 5.5 (Hessian identity)**.** *Write $H_{\boxplus_n}^{(i)} := \operatorname{Hess} \Omega_{\boxplus_n, i}$ for the Hessian of the i-th coordinate function of $\Omega_{\boxplus_n}$ at (α, β). Then for all $w = (u, v) \in V$:*

$$w^T J_{\boxplus_n} J_{\boxplus_n}^T w \;=\; \|w\|^2 - \sum_{i=1}^n \gamma_i\, w^T H_{\boxplus_n}^{(i)}\, w. \tag{10}$$

Proof. Fix $w = (u, v) \in V$ and consider the paths $\alpha(t) := \alpha + tu$, $\beta(t) := \beta + tv$, $\gamma(t) := \Omega_{\boxplus_n}(\alpha(t), \beta(t))$. By variance additivity (Lemma 3.7) applied to root vectors:

$$\frac{1}{n} \sum_i \gamma_i(t)^2 - \left(\frac{1}{n} \sum_i \gamma_i(t)\right)^2 = \operatorname{Var}(\alpha(t)) + \operatorname{Var}(\beta(t)).$$

Since $(u, v) \in V$, the means $\frac{1}{n} \sum \alpha_i(t)$ and $\frac{1}{n} \sum \beta_i(t)$ (and hence $\frac{1}{n} \sum \gamma_i(t)$) are constant in t. Differentiating the equation $\frac{1}{n} \sum_i \gamma_i(t)^2 = \operatorname{Var}(\alpha(t)) + \operatorname{Var}(\beta(t)) + \text{const}$ twice at $t = 0$:

$$\frac{1}{n} \sum_i \left[(D_w \gamma_i)^2 + \gamma_i\, D_w^2 \gamma_i\right] = \frac{1}{n}(\|u\|^2 + \|v\|^2),$$

since $\partial_t^2[\|\alpha + tu\|^2] = 2\|u\|^2$ and similarly for v. Now $\sum_i (D_w\gamma_i)^2 = w^T J_{\boxplus_n} J_{\boxplus_n}^T w$ and $D_w^2\gamma_i = w^T H_{\boxplus_n}^{(i)} w$, giving (10). □

Lemma 5.6 (PSD Hessian sum)**.** *For any $c_1 \geq \cdots \geq c_n$, the matrix $\sum_{i=1}^n c_i\, H_{\boxplus_n}^{(i)}$ is positive semi-definite.*

Proof. Consider the multivariate polynomial

$$f(x, a_1, \ldots, a_n, b_1, \ldots, b_n) := \mathop{\mathbb{E}}_{\pi \in S_n} \prod_{i=1}^n (x - a_i - b_{\pi(i)}),$$

which is homogeneous and, since $\boxplus_n$ preserves real-rootedness, hyperbolic in the direction $e_1 = (1, 0, \ldots, 0)$. Its roots in x (at fixed $a = \alpha$, $b = \beta$) are the entries γ_i of $\Omega_{\boxplus_n}(\alpha, \beta)$. By a theorem of Bauschke, Güler, Lewis, and Sendov [1], the partial sums $\sigma_k(a, b) := \sum_{i=1}^k \gamma_i(a, b)$ are convex functions; in particular, $\operatorname{Hess} \sigma_k \succeq 0$. Since the c_i are decreasing, $\sum c_i\gamma_i$ is a positive linear combination of the σ_k, hence $\operatorname{Hess}(\sum c_i\gamma_i) = \sum c_i H_{\boxplus_n}^{(i)} \succeq 0$. □

Proof of Proposition 5.4. For $(u, v) \in V$, by Lemma 5.5:

$$\|J_{\boxplus_n}(u, v)\|^2 = \|u\|^2 + \|v\|^2 - \sum_{i=1}^n \gamma_i\, w^T H_{\boxplus_n}^{(i)}\, w.$$

The roots $\gamma_1 \geq \cdots \geq \gamma_n$ are ordered, so Lemma 5.6 (with $c_i = \gamma_i$) gives $\sum \gamma_i\, w^T H_{\boxplus_n}^{(i)}\, w \geq 0$. □

5.4. **Main theorem.**

Theorem 5.7 (Finite free Stam inequality)**.** *For all $n \geq 2$ and $p, q \in \mathcal{P}_n^{\mathbb{R}}$:*

$$\frac{1}{\Phi_n(p \boxplus_n q)} \;\geq\; \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}. \tag{11}$$

Proof. By Lemma 5.2, we may assume p, q, and $r := p \boxplus_n q$ all have simple roots. Write α, β, γ for their ordered root vectors.

Step 1: Score vectors are in V. By Lemma 3.3(i), $\sum_i V_i(\alpha) = 0$ and $\sum_i V_i(\beta) = 0$, so $(V(\alpha), V(\beta)) \in V$.

Step 2: Combine contraction with score–Jacobian relation. Proposition 5.4 with $u = aV(\alpha)$ and $v = bV(\beta)$ (for $a, b > 0$ to be chosen) gives

$$\big\|J_{\boxplus_n}(aV(\alpha), bV(\beta))\big\|^2 \leq a^2\, \Phi_n(p) + b^2\, \Phi_n(q).$$

By Proposition 5.3, the left-hand side equals $(a + b)^2\, \Phi_n(r)$. Hence

$$(a + b)^2\, \Phi_n(r) \;\leq\; a^2\, \Phi_n(p) + b^2\, \Phi_n(q). \tag{12}$$

Step 3: Blachman's choice. Set $a = 1/\Phi_n(p)$ and $b = 1/\Phi_n(q)$. Then (12) becomes

$$\Big(\frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}\Big)^2 \Phi_n(r) \;\leq\; \frac{1}{\Phi_n(p)} + \frac{1}{\Phi_n(q)}.$$

Dividing both sides by $\big(1/\Phi_n(p) + 1/\Phi_n(q)\big)\Phi_n(r) > 0$ yields (11). □

6. Discussion

The finite free Stam inequality (Theorem 5.7) is now established for all $n \geq 2$. The proof combines the commutation of $\boxplus_n$ with differential operators, variance additivity, and the convexity theory of hyperbolic polynomials. The structural identities developed in Section 3 (Fisher–repulsion, Laplacian, Bezoutian) and the Hermite heat equation (Theorem 3.9) provide complementary perspectives. The independent proofs for $n \leq 3$ (Theorem 4.1) and for Gaussian inputs (Theorem 4.3) use different techniques (sum-of-squares and dissipation, respectively) and yield additional information such as exact defect formulas.

Remark 6.1 (Equality characterisation). For $n = 3$, Theorem 4.1 shows that equality in the Stam inequality holds (for inputs with $u_p, u_q > 0$) if and only if $\ell_3(p) = \ell_3(q) = 0$, i.e., both inputs are finite Gaussians. The characterisation of equality for general n remains an interesting open question.

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